Learning and the Efficiency of Expert Referrals

SUPPLEMENTAL APPENDIX

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November 2026

1 Data and Descriptive Statistics

This section provides additional details of our analytic data and subsequent referral patterns.

1.1 PCP Identification

Our method identifies a patient's PCP using a carrier-claims lookback window, selecting the most-visited PCP in pre-surgery E&M claims. Table 1 compares this approach with the referring physician listed on the surgeon's claim (`rfr_physn_npi`). The listed referring physician field is not populated in roughly 32% of cases, and when it is populated, it most often lists the surgeon (self-referral, about 43%) or a non-PCP specialist (about 7%). Only about 18% of cases list a PCP as the referring physician. Among cases where both methods identify a PCP, concordance exceeds 82% in every year.

Figure 1 shows the composition of the listed referring physician field alongside our lookback coverage rate. Our lookback method identifies a PCP for approximately 80% of procedures, providing substantially better coverage than the listed field alone.

2 Construction of the Referral-Order Event Study

Every referral in our sample defines a window. The window consists of the referral itself, the PCP's five referrals before it, and the PCP's five referrals after it, indexed by $t = -5, \ldots, 5$

with the defining referral at $t = 0$. The specialist of the window is the one who received the referral at $t = 0$, and the outcome at each other position is whether that referral went to the same specialist. A window is a failure window if the surgery at $t = 0$ was a failure and a success window otherwise. The event study in the main text compares the change in the outcome from the pre-period to the post-period between failure windows and success windows of the same PCP and of the same specialist and year.

Table 7 illustrates with a PCP whose 22 referrals rotate among three specialists. The 22 referrals define 22 windows, seven for specialist 1 (centered on referrals 1, 4, 9, 11, 14, 17, and 20), seven for specialist 2, and eight for specialist 3, and the table shows three of them. Referral 6 goes to specialist 2 and fails. Every other referral succeeds. After the failure the PCP avoids specialist 2 for several referrals, so referrals 7 through 11 go to specialists 1 and 3, including referral 8, which would otherwise have gone to specialist 2, and the PCP first returns to specialist 2 at referral 13. The window centered on referral 6 is a failure window for specialist 2. Two of the five referrals in its pre-period went to specialist 2 and none of the five in its post-period did, and this change is the response that the event study measures.

Because every referral defines a window, each referral also lies in the windows of its neighbors, at a different position in each and compared with that window's specialist. Referral 8 is at position $+2$ in the window centered on referral 6, where the question is whether it went to specialist 2, and at position -4 in the window centered on referral 12, where the question is whether it went to specialist 3. The rows for one referral are therefore one choice recorded several times, and a referral that a failure diverted is a zero in the failure window and a one in a window for the specialist who received it instead.

We keep a window only if it contains no other failure. In the example, referral 6 lies in the windows centered on referrals 1 through 5 and 7 through 11, so all ten are dropped. These include the specialist 1 windows centered on referrals 1, 4, 9, and 11. The windows kept are the failure window and the success windows centered on referrals 12 through 22. The first success window kept after the failure is the one centered on referral 12, for specialist 3. This rule is blunt. It drops windows in which the failure sits at the edge and nothing in the window could have been affected, and in our data it keeps 466,954 of 1,190,521 referrals as windows. Because the PCP fixed effects are interacted with position, a PCP contributes to the estimates only with at least one failure window and at least one success window.

The rule does not remove every window whose outcomes reflect the response. The pre-period of the window centered on referral 12 is referrals 7 through 11, the post-period of the failure window. Referral 8 is a zero in the failure window, because it did not go to specialist 2, and a one in the success window, because it went to specialist 3 instead, and both entries record the same response. The pre-period of the success window is elevated, three of five against one of five in its post-period, so its change of $-2/5$ matches the change in the failure window and the comparison between the two is zero. The success windows centered on referrals 13 through 16 have the same problem.

We therefore also estimate the event study after dropping any window that overlaps the post-period of another failure's window. In the example, the post-period of the failure window is referrals 7 through 11, and the windows centered on referrals 12 through 16 each contain some of them, so this drops those five windows, leaving the failure window and the success windows centered on referrals 17 through 22. The first of these windows is centered on referral 17, for specialist 1. Its pre-period and post-period each contain one referral to specialist 1, so its change is zero and the comparison recovers the $-2/5$ response. After this restriction no referral made while the PCP was responding to a failure appears in any success window. The dependence that remains across a PCP's windows involves referrals that no failure moved, and we account for it by clustering standard errors by PCP. In the data the overlap offsets only part of the response, and the restriction moves the estimate at the first referral after a failure from -0.0115 to -0.0146.

3 Identification

We formally show identification of the myopic model below. Identification of the forward-looking model, discussed in the main paper, follows because the Brezzi-Lobo approximation to the Gittins index involves the same free parameters (ρ and η) on the same data as the myopic mean belief, and the present discounted value of familiarity has a known relationship with its current value. The forward-looking model therefore introduces no additional free parameters beyond those of the myopic model.

3.1 Myopic Model

We start by proving the identification of the myopic model. To review (and simplify), the utility to PCP i for referring patient t to specialist j is

$$U_{ijt} \equiv \alpha m_{ijt} + f(e_{ijt}) + \xi_j + \epsilon_{ijt},$$

where m_{ijt} is the current mean of the beliefs about the probability of success at specialist j (using the history before patient t), e_{ijt} is the amount of experience with specialist j (the number of past patients i sent to j), $f(e_{ijt})$ is the subjective utility from familiarity with specialist j, ξ_j is an unobservable specialist fixed effect, and ϵ_{ijt} is an idiosyncratic shock with a known distribution. Also let $\check{y}_{ijt}$ denote the number of successes among past patients that i sent to j.

Given a parametric assumption on the error term, ϵ, the observable choice probabilities directly identify differences in utility, excluding the error term: e.g., $(U_{ijt} - \epsilon_{ijt}) - (U_{ikt} - \epsilon_{ikt})$. To simplify these expressions, let v denote utility without the error term: $v \equiv U - \epsilon$.

To be somewhat more precise, there is a finite number of specialists but large numbers of patients and PCPs. The choice probabilities are

$$\Pr(j | (e_{ikt}, \check{y}_{ikt})_{k=1}^{J}),$$

where "j" means the event that specialist j is chosen. We may need consistent estimates of the choice probabilities for cases where $e_{ikt} = 0, \forall k$, and where $e_{ikt} \to \infty, \forall k$. Hence the number of PCPs may need to go to infinity, but the number of patients must grow faster.

We identify the remaining components of v in steps as follows.

1. Specialist fixed effects (ξ): Any cases where the observables are the same for two specialists can be used to identify the fixed effects (up to a standard normalization; e.g., one fixed effect is set to zero). Because if $e, \check{y}, x, n$ are the same, then

$$v_{ijt} - v_{ikt} = \xi_j - \xi_k.$$

   In particular, cases where $e = 0$ (and hence $\check{y} = 0$) are useful here.

2. Altruism parameter (α): This uses cases with large past experience, formally the limit when $e \to \infty$. Let $e_{ijt} \to \infty$ and $e_{ikt} \to \infty$ (and for convenience $x_{ijt} = x_{ikt}$ and $n_j = n_k$). Then

$$v_{ijt} - v_{ikt} = \frac{\alpha \bar{y}_{ijt}}{0+1} + \xi_j - \frac{\alpha \bar{y}_{ikt}}{0+1} - \xi_k.$$

Hence

$$\alpha = \frac{(v_{ijt} - \xi_j) - (v_{ikt} - \xi_k)}{\bar{y}_{ijt} - \bar{y}_{ikt}}$$

in these limits. Thus cases where a PCP has large past experience with two specialists, but different success rates with them, identify the altruism parameter.

3. Strength of priors ($a_0 + b_0$): The responsiveness to success with finite experience then identifies the strength of priors (i.e., the stickiness of beliefs). Let $e_{ijt} = e_{ikt} = e$ (and for convenience $x_{ijt} = x_{ikt}$ and $n_j = n_k$). Then

$$v_{ijt} - v_{ikt} = \frac{\alpha \check{y}_{ijt}}{a_0 + b_0 + e} + \xi_j - \frac{\alpha \check{y}_{ikt}}{a_0 + b_0 + e} - \xi_k.$$

Hence

$$(a_0 + b_0) = \frac{\alpha(\check{y}_{ijt} - \check{y}_{ikt})}{(v_{ijt} - \xi_j) - (v_{ikt} - \xi_k)} - e,$$

and all the objects on the right are observable or already identified.

4. Effect of familiarity ($f(e)$): Finally, differences in experience with specialists identify the effect of familiarity. For convenience let x, n be the same, then

$$v_{ijt} - v_{ikt} = \alpha \frac{a_0 + \check{y}_{ijt}}{a_0 + b_0 + e_{ijt}} + f(e_{ijt}) + \xi_j - \alpha \frac{a_0 + \check{y}_{ikt}}{a_0 + b_0 + e_{ikt}} - f(e_{ikt}) - \xi_k.$$

Hence

$$\begin{aligned} f(e_{ijt}) - f(e_{ikt}) \; &= (v_{ijt} - \xi_j) - (v_{ikt} - \xi_k) \\ &\quad - \alpha \left[\frac{a_0}{a_0 + b_0 + e_{ijt}} - \frac{a_0}{a_0 + b_0 + e_{ikt}} \right] \\ &\quad - \alpha \left[\frac{\check{y}_{ijt}}{a_0 + b_0 + e_{ijt}} - \frac{\check{y}_{ikt}}{a_0 + b_0 + e_{ikt}} \right]. \end{aligned}$$

This identifies f up to a normalization, specifically $f(0) = 0$.

However this also requires the value of a_0, which can be obtained if the prior ratio, $\frac{a_0}{a_0+b_0}$, is calibrated separately. Intuitively, we need to fully identify the priors (both the ratio and the strength) in order to recover the effect of familiarity, because we need to know exactly how experience washes away the priors in order to separate that from the effect of familiarity.

3.2 Monte Carlo Evidence on the Recovery of α

To assess the finite-sample properties of our estimator, we conduct a Monte Carlo simulation calibrated to match key features of our data. We generate referral markets with 50 PCPs and 10 specialists, where each PCP makes 15 referrals per year. We include a burn-in period to establish referral histories and rely on a 6-year estimation window to match the estimation in our real data. Specialists are assigned heterogeneous quality levels (true failure probabilities) drawn from a normal distribution centered at the base failure rate, and each specialist receives a fixed effect drawn independently. PCPs choose specialists according to the myopic structural model presented in the main text, with beliefs updated via Bayes' rule after each patient outcome.

We estimate $\hat{\alpha}$ using the same constrained maximum likelihood procedure used in the main estimation (see Section 4), with specialist fixed effects included in the multinomial logit. We vary the true α over the range 0.05 to 0.60, the base failure rate over 0.05 to 0.15, and the quality dispersion (standard deviation of failure rates across specialists) over 0.01 to 0.05, yielding 77 parameter configurations after excluding cases where the quality dispersion exceeds the base failure rate. For each configuration, we run 50 replications.

Figure 2 presents the bias in $\hat{\alpha}$ as a function of the true α and the base failure rate, holding quality dispersion fixed at 0.03 (approximately matching the interquartile range of failure rates in the data). The surface reveals that bias is primarily driven by the magnitude of the true α, with a secondary role for the base failure rate. Bias is largest when the true α is small, because the signal from patient outcomes is weak relative to the noise from the logit errors and the specialist fixed effects. Higher base failure rates improve identification by producing more frequent failures, which accelerates belief updating.

Figure 3 presents the relative bias (as a percentage of the true α) for each failure rate,

again at quality dispersion of 0.03. Relative bias falls below 10% for $\hat{\alpha} \geq 0.30$ when the base failure rate is 0.09 or higher. Among markets with non-zero estimates of $\hat{\alpha}$, the mean is 0.25 for $\eta = 1$ and 0.43 for $\eta = 5$ in the myopic model (0.32 and 0.54 in the forward-looking model), with the 75th percentile at 0.32 and above in all cases. The overwhelming majority of markets therefore either suggest very little learning ($\hat{\alpha} = 0$) or fall in the range where the estimator appears to perform well. These simulation results also support our decision to fix η rather than estimate it jointly with α, as the likelihood surface with respect to η is relatively flat across the parameter configurations we examined.

3.3 Simulation study on identification of η

Although we have shown that η is identified, our preliminary estimates of this parameter exhibited poor performance and so we used fixed values instead. Here we present a simulation study that indicates the data in our application would indeed have little power to inform an estimate of η.

We simulate a market with 2,500 patients and 25 specialists. For simplicity, the PCP-specialist histories (# referrals, # successes) are independent across patients, as though each patient were treated by a different PCP, but the PCPs have prior histories with some specialists. Each PCP has a 0.2 probability of any history with each specialist, independent across specialists. Conditional on any history with a specialist, the number of past referrals is Poisson with mean 4, and the number of past successes is binomial using the number of past referrals and the surgeon's success probability (explained below).

The simulation model has the same specification as our empirical model, except that there is no congestion effect and the familiarity effect is specified as $f(e_{ijt}) = \beta \ln(1 + e_{ijt})$. The parameter values are as follows:

- $\alpha = 0.3$ (weight on outcome)
- $\pi = -0.07$ (weight on distance)
- $\beta = 0.9$ (weight on familiarity)
- $\eta = 3$ (prior strength)
- ρ: explained below (prior mean)
- ξ: explained below (fixed effects)

Specialist qualities, q_j, are drawn independently from a normal distribution with mean 0.9 and SD 0.04, and censored symmetrically at 0.81 and 0.99. The prior mean, ρ, is set equal to the average of these draws. The specialist fixed effects, ξ_j, are given fixed values on a grid from -1.5 to $+1.5$ with equal intervals of $3/25$ between values.

Figure 4 shows the results from four representative simulations. It plots log-likelihood values obtained using a wide range of values for η while keeping the other parameters at their true values. It is apparent that the likelihood function is very flat in this parameter. The log likelihood changes by only a few points over the entire span of values for η. Across 100 such simulations, the average range of the likelihood values is only 2.1 log points and the 95th percentile is 4.3 log points. Based on this, we do not think the data in our application would have good power to yield an informative estimate of η.

4 Estimation

4.1 Iterative Estimation of α

To constrain $\alpha \geq 0$ we estimate $\ln \alpha$ and specify the beliefs term as $\exp(\ln \alpha) \cdot m$ (or $\exp(\ln \alpha) \cdot g(m, v)$ in the forward-looking model). We use an iterative procedure proposed in Fader et al. (1992) which makes it possible to employ standard estimation commands designed for multinomial logits where all terms are linear in parameters. The procedure is based on a first-order Taylor series approximation of the term that is nonlinear in parameters. With the exponential function this is simply

$$\exp(\ln \alpha_2) \cdot m \approx [\exp(\ln \alpha_1) + (\ln \alpha_2 - \ln \alpha_1) \exp(\ln \alpha_1)] \cdot m,$$

(because $d \exp(\ln \alpha)/d \ln \alpha = \exp(\ln \alpha)$), where $\ln \alpha_1$ and $\ln \alpha_2$ are two candidate values.

In our case, the iteration works by setting a value of $\ln \alpha$ (e.g., $\ln \alpha_1$) and then estimating the model with a free coefficient (e.g., $\hat{\pi}_1$) on the term $\exp(\ln \alpha_1) \cdot m$. Rearranging the Taylor approximation above shows how the estimated coefficient can be used to update the value

of $\ln \alpha$, as follows:

$$\exp(\ln \alpha_2) \cdot m \approx \underbrace{[1 + (\ln \alpha_2 - \ln \alpha_1)]}_{\hat{\pi}_1} \exp(\ln \alpha_1) \cdot m;$$

hence:

$$\ln \alpha_2 = \ln \alpha_1 + \hat{\pi}_1 - 1.$$

Furthermore, because the estimated coefficient $\hat{\pi}_1$ by definition improves the likelihood, using $\exp(\ln \alpha_2)$ should produce a higher likelihood than $\exp(\ln \alpha_1)$. Fader et al. (1992) prove that this algorithm converges to the MLE of $\ln \alpha$ under standard regularity conditions. For some intuition note that at the maximum, $\ln \alpha_*$, the likelihood cannot be improved, and so the value of $\hat{\pi}$ should be one; i.e., in terms of the Taylor expansion

$$\exp(\ln \alpha_*) = \underbrace{[1 + (\ln \alpha_* - \ln \alpha_*)]}_{\hat{\pi}_*} \exp(\ln \alpha_*) = \underbrace{[1]}_{\hat{\pi}_*} \exp(\ln \alpha_*)$$

The standard error then follows from Fader et al. (1992) as

$$SE(\hat{\alpha}) = \exp(\ln \alpha_*) SE(\hat{\pi}_*)$$

because $\hat{\alpha} = \exp(\ln \alpha_*) = \exp(\ln \alpha_* + \hat{\pi}_* - 1)$ and asymptotically we only need to consider the variation in $\hat{\pi}_*$ (see Lemma 2 in Fader et al., 1992).

4.2 First-Stage Diagnostics for Congestion

Specialist patient volume, n_{jt}, is endogenous in the referral utility because unobserved specialist characteristics that raise utility (and therefore referrals) would mechanically raise volume. To separate the causal effect of congestion on referral appeal from this correlation, we instrument for n_{jt} using a distance-based predicted patient count, following Richards-Shubik et al. (2022). For each specialist and time period, the instrument is a distance-predicted patient count, constructed by estimating a conditional logit of referral choice with patient-to-surgeon distance as the sole covariate, then summing the predicted choice probabilities across patients for each specialist. The instrument varies across specialists through

the geographic distribution of PCPs and patients, and it is independent of specialists' unobserved quality or taste shifters conditional on the HRR fixed effects that absorb market-wide demand. The exclusion restriction is standard in the spatial choice literature: conditional on HRR, the geographic configuration of PCPs relative to specialists affects referrals only through the distance channel, not directly through quality.

The first stage regresses specialist total patients on the distance-based instrument, HRR fixed effects, and a time-period indicator, weighted by the inverse variance of the specialist fixed effect estimate and clustered at the specialist level. Table 2 reports the pooled results. The first-stage coefficient is close to one and precisely estimated, indicating that a one-patient increase in the distance-predicted count corresponds to roughly a one-patient increase in realized volume after absorbing market-level differences. The F-statistic on the excluded instrument exceeds one thousand in both specifications, well above conventional thresholds for weak-instrument concerns (Stock and Yogo, 2005). The estimates are nearly identical across $\eta = 1$ and $\eta = 5$ because the instrument and the endogenous variable do not depend on the learning model.

Figure 5 presents partial scatter plots after residualizing both the instrument and patient volume on HRR fixed effects and the time-period indicator. The relationship is visibly linear across the range of the residualized instrument, with no evidence of influential observations driving the pooled coefficient. Weighting by the inverse variance of the specialist fixed effect gives more influence to specialists with more precisely estimated effects, which also tend to be higher-volume specialists for whom the congestion mechanism is most empirically relevant.

A pooled first stage averages across HRRs of different sizes, so we also estimate the first stage separately in each HRR. Figure 6 shows the distribution of HRR-level F-statistics, weighted by the number of patients in each HRR so that larger markets (where most identification comes from) receive more weight. The distribution is concentrated well above the conventional $F = 10$ threshold, with the large majority of patients in markets that comfortably satisfy the weak-instrument diagnostic. A minority of smaller HRRs exhibit low F-statistics, reflecting the small number of specialists and the limited spatial variation within those markets rather than a systematic failure of the instrument. Because our structural estimation pools information across HRRs through the common γ, the strength of the pooled first stage is the relevant benchmark for inference on congestion.

4.3 Model Fit Diagnostics

We examine two diagnostics that together assess model fit and defend the structural interpretation of the parameters. Figure 7 plots predicted versus actual referral shares at the specialist level, which is a standard validation for discrete-choice models. The predicted shares track actual shares closely across the range, with no systematic bias at the high or low end.

Figure 8 addresses a different concern. If the specialist fixed effects, ξ_j, were simply absorbing quality differences that our measure does not capture, then the estimated α would be biased toward zero and the reported counterfactuals would understate the gains from information. The figure shows that the within-HRR correlation between estimated ξ_j and observed specialist quality is near zero in most markets. The fixed effects therefore capture demand factors that are empirically distinct from outcome-based quality, supporting the interpretation of α as a separate learning channel.

4.4 Sensitivity of the Quality Response to Alternative Specifications

Re-estimating the structural model and recomputing counterfactuals for every alternative specification is computationally prohibitive. As a more tractable diagnostic, we examine how the estimated quality response, α, varies across alternative specifications of the choice model, estimated separately within each HRR. In the myopic case, the choice model is

$$\mathrm{E}\left[U_{ijt} \mid \dots\right] = \alpha\, m_{ijt} + \pi x_{ijt} + \sum_{b \in B} \kappa_b\, I(e_{ijt} \in b) + \delta_j + \epsilon_{ijt}, \tag{1}$$

where m_{ijt} is the posterior mean belief about specialist j's quality and δ_j is a single, time-invariant specialist fixed effect. This matches the myopic specification in the main text, except that the composite time-varying effect $\delta_{j,\tau(t)} = \gamma n_{j,\tau(t)} + \xi_j$ is replaced by one fixed effect δ_j per surgeon, which lets the diagnostic run quickly across the full set of HRRs. We estimate α with the same iterative procedure used for the structural model, which constrains $\alpha \geq 0$. Because the counterfactual reallocations and health effects scale directly with α in each market, stability of α across specifications implies stability of our counterfactual

conclusions.

Table 3 reports the distribution of $\hat{\alpha}$ across HRRs under three specifications. The "Baseline" specification matches the main-text estimates. The "Integrated" specification adds an indicator for the PCP and specialist sharing a practice. The "Peer Outcomes" specification further includes a measure of specialist quality constructed from the outcomes of other PCPs' patients in the same practice. All three impose time-invariant congestion for computational ease. The mean $\hat{\alpha}$ moves from 0.192 in the baseline to 0.192 with the practice indicator and 0.203 with the peer-quality measure, and the upper-tail percentiles are similarly stable. The estimated quality response is therefore not driven by failing to control for practice integration or peer-based quality information.

4.5 Risk Adjustment

Here, we discuss how risk adjustment does not materially affect our estimates of specialist quality. First, our sample is heavily concentrated in a single, low-complexity DRG. Table 4 reports the distribution of admissions across DRGs. DRG 470 (major joint replacement of lower extremity, without MCC) accounts for 86.1 percent of the 2.9 million events, with the remaining 14 percent spread across 17 procedure codes. The combination of elective admission and physician referral as sample inclusion criteria selects a planned, lower-complication population in which case-mix variation across specialists is limited. Indeed, consistent with this homogeneity, the Comprehensive Care for Joint Replacement model similarly incorporated limited additional patient-level risk adjustment in setting its bundled payment amounts (Group, 2021).

Second, raw and risk-adjusted specialist quality are highly correlated. We construct a risk-adjusted quality measure by regressing the patient-level 90-day failure indicator on age (quadratic), sex, race, DRG, and the Charlson Comorbidity Index, then taking each specialist's observed minus expected failure rate plus the overall sample mean. The Pearson correlation between the raw and risk-adjusted measures is 0.84, so adjusting for observable patient case-mix moves the cross-section of specialist quality very little on average.

Third, the estimated quality response is invariant to either case-mix sensitivity. Table 5 reports the distribution of $\hat{\alpha}$ under two re-estimations of the choice model in Equation (1).

The first restricts the estimation sample to DRG 470 only, and the second swaps the risk-adjusted specialist-quality measure into the belief prior. The mean of $\hat{\alpha}$ moves from 0.192 in the baseline (Table 3) to 0.216 when restricting to DRG 470 only, and to 0.192 with the risk-adjusted quality measure. The upper-tail percentiles are similarly stable.

Because the sample is concentrated in a single elective DRG, the raw and risk-adjusted quality measures are tightly correlated, and the estimated quality response is invariant to either case-mix sensitivity, we retain the raw specialist-quality measure throughout. Even setting these results aside, the structural model would not easily accommodate a continuous risk-adjusted signal. The beta-binomial belief-updating framework presumes binary outcomes, and the path to a comparable model that admits continuous signals (e.g., normal-normal updating with a corresponding Gittins index) is unclear.

5 Counterfactual Robustness

5.1 Counterfactuals without Familiarity

The purpose of this counterfactual is to estimate the potential changes in referrals and subsequently patient health if PCPs could rely entirely on specialist quality, with full information, and without any attachment due to prior relationships. These results are presented in Figure 9. Here, we estimate larger effects on referral probabilities, with reallocation of 35.46 percent on average across HRRs. However, the health effects of this scenario are negative on average, with a change in success rates of -0.27 percentage points. Without any familiarity to anchor referral patterns, and with the estimated weight on quality small relative to distance preferences and specialist fixed effects, the reallocation proceeds largely independently of quality. This result reinforces the finding in the main text that existing referral relationships encode quality information, and that removing them eliminates an important source of quality sorting in the referral market.

5.2 Equilibrium Robustness

To confirm the robustness of the counterfactual equilibrium, we re-solve the fixed point from two extreme starting values, zero patients at every specialist and the observed patient

volumes, in a stratified sample of six HRRs spanning small, medium, and large markets. Table 6 reports the results. Across 1,021 specialist-period observations, the mean absolute difference in specialist shares within an HRR-period is 0.7 to 1.9 percentage points, and roughly three-quarters of specialists agree to within one percentage point across the two starts. At the HRR level, the patient-weighted mean of specialist quality differs by at most 7×10^{-5} between the two starting values, roughly an order of magnitude smaller than the counterfactual-induced changes we report. Our counterfactual equilibrium is therefore robust to starting value, and the aggregate outcomes we use in the analysis are effectively invariant.

5.3 Distributional Effects on Quality

In addition to the changes in mean quality reported in the main text, we examine how counterfactual reallocation reshapes the within-HRR distribution of specialist quality. Figures 10 and 11 present off-diagonal scatters of the 10th and 90th patient-weighted percentiles of specialist quality under each counterfactual against the corresponding baseline values. Points falling on the 45-degree line represent HRRs where the percentile is unchanged, which dominates in most markets because weighted percentiles are step functions over a discrete set of specialists and small reallocations do not cross the relevant specialist boundary. The off-diagonal scatters drop these unchanged points to focus attention on HRRs where the distribution actually shifts.

Two patterns emerge. Under the existing-familiarity counterfactual, both the 10th and 90th percentiles of quality shift upward on average in markets where they move, consistent with the distribution compressing toward higher quality and producing the small positive mean quality effect reported in the main text. Under the reset-familiarity counterfactual, the 10th percentile shifts downward while the 90th shifts upward, indicating that resetting familiarity increases dispersion. A minority of patients are reallocated to higher-quality specialists they had previously been far from, while another group loses the quality cues embedded in their prior referral patterns and is reallocated to lower-quality specialists. The net mean effect under the reset-familiarity counterfactual is slightly negative because the losses at the bottom of the distribution outweigh the gains at the top.

5.4 Distributional Effects on Spending

As a diagnostic on the direction of these reallocations, we also examine how the counterfactual shifts the within-HRR distribution of episode spending. Figures 12 and 13 present off-diagonal scatters of the 10th and 90th patient-weighted percentiles of episode spending under each counterfactual. The broad widening pattern of the quality distribution under reset familiarity also appears in spending, with the 10th and 90th percentiles moving in opposite directions in markets where the distribution shifts. Some of this co-movement between spending and quality is mechanical, since surgical complications and readmissions enter both our quality measure and 90-day episode spending.

References

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Tables and Figures

Table 1: PCP Identification Method Validation[a]

		Composition of Referral Field (%)				Lookback	
Year	Procedures	Not Pop.	Self-Referral	Non-PCP	PCP	Coverage (%)	Concordance (%)
2013	445,274	30.1	41.6	7.5	20.8	81.5	82.8
2014	456,377	31.2	41.6	7.1	20.1	80.9	82.7
2015	502,927	31.1	42.9	7.1	18.9	80.5	82.8
2016	542,126	31.5	43.7	7.1	17.8	79.8	82.4
2017	557,277	32.6	44.1	6.7	16.6	79.0	82.7
2018	506,138	34.4	43.5	6.5	15.6	78.2	82.1
All	3,010,119	31.9	43.0	7.0	18.2	79.9	82.6

[a]Comparison of our carrier-lookback PCP identification method with the referring physician listed on the surgeon's claim (`rfr_physn_npi`). "Not Pop." indicates the referral field is not populated. "Self-Referral" indicates the listed referring physician is the operating surgeon. "Non-PCP" indicates a non-PCP specialist. "PCP" indicates a primary care physician. "Lookback Coverage" is the share of procedures for which our method identifies a PCP. "Concordance" is the share of cases where both methods identify a PCP and they agree.

Table 2: First-Stage Results for Congestion 2SLS[a]

	Coefficient	SE	F-statistic	N
$\eta = 1$	1.418	(0.040)	1236.5	36,509
$\eta = 5$	1.415	(0.041)	1171.9	36,543

[a]Pooled first-stage regression of specialist total patients on distance-predicted patient counts, with HRR fixed effects and a time-period indicator. Observations weighted by the inverse variance of the specialist fixed effect estimate. Standard errors clustered at the specialist level. Results shown for the myopic model; the forward-looking model yields nearly identical first-stage estimates because the instrument and endogenous variable do not depend on the learning model.

Table 3: Sensitivity Analysis[a]

		Percentile				
Parameter	Mean	10th	25th	50th	75th	90th
Baseline	0.1924	0.0000	0.0000	0.1063	0.2766	0.5499
Integrated	0.1921	0.0000	0.0000	0.1122	0.2785	0.5015
Peer Outcomes	0.2033	0.0000	0.0000	0.1137	0.3132	0.5811

[a]Estimates of α (for $\eta = 1$) across alternative specifications. "Baseline" denotes our original estimates for the myopic learning model presented in the main text; "Integrated" denotes the estimates for α when including a dummy variable for whether the PCP and specialist are part of the same practice; and "Peer Outcomes" denotes the estimates of α when also including a measure of specialist quality based on the PCP's peers (e.g., another measure of quality based solely on the outcomes of other PCPs' patients in the same practice). All specifications are otherwise identical, where we also impose time-invariant congestion for computational ease.

Table 4: Distribution of DRG Codes[a]

DRG	Description	Events	Share
470	Major joint replacement of lower extremity, without MCC	2,518,132	0.8611
483	Major joint and limb reattachment of upper extremity	168,168	0.0575
469	Major joint replacement of lower extremity, with MCC	81,318	0.0278
462	Bilateral or multiple major joint procedures of lower extremity, without MCC	56,731	0.0194
481	Hip and femur procedures except major joint, with CC	23,083	0.0079
489	Knee procedures without principal diagnosis of infection, without MCC	17,984	0.0061
482	Hip and femur procedures except major joint, without CC/MCC	13,702	0.0047
488	Knee procedures without principal diagnosis of infection, with MCC	11,482	0.0039
512	Shoulder, elbow, or forearm procedures, except major joint, without CC/MCC	8,371	0.0029
480	Hip and femur procedures except major joint, with MCC	5,822	0.0020
486	Knee procedures with principal diagnosis of infection, with CC	5,385	0.0018
511	Shoulder, elbow, or forearm procedures, except major joint, with CC	3,046	0.0010
461	Bilateral or multiple major joint procedures of lower extremity, with MCC	2,848	0.0010
487	Knee procedures with principal diagnosis of infection, without CC/MCC	2,613	0.0009
508	Major shoulder or elbow joint procedures, without CC/MCC	2,571	0.0009
507	Major shoulder or elbow joint procedures, with CC/MCC	1,305	0.0004
485	Knee procedures with principal diagnosis of infection, with MCC	1,097	0.0004
510	Shoulder, elbow, or forearm procedures, except major joint, with MCC	733	0.0003
Total		2,924,391	1.0000

[a]Distribution of admissions across DRG codes in the sample (2008–2018, major joint replacement procedures with elective admission and physician referral). "Events" is the number of admissions in each DRG; "Share" is the share of all admissions.

Table 5: Sensitivity to Case-Mix Adjustments[a]

		Percentile				
Parameter	Mean	10th	25th	50th	75th	90th
DRG 470 only	0.2157	0.0000	0.0000	0.1305	0.3137	0.5718
Risk-Adjusted Quality	0.1921	0.0000	0.0000	0.1174	0.2815	0.4774

[a]Estimates of α (for $\eta = 1$) under two case-mix sensitivities of the first-step multinomial logit. "DRG 470 only" restricts the estimation sample to admissions in DRG 470 (major joint replacement of lower extremity, without MCC). "Risk-Adjusted Quality" replaces the raw specialist-quality measure in the belief prior with a risk-adjusted measure constructed via indirect standardization of the patient-level 90-day failure indicator on age (quadratic), sex, race, DRG, and the Charlson Comorbidity Index. Compare to the baseline distribution in Table 3.

Table 6: Equilibrium Robustness to Starting Values[a]

HRR	Per	N spec	Mean $\lvert\Delta p_{ij}\rvert$	Share <1pp	$\bar{q}_0$	$\bar{q}_1$	$\bar{q}_1 - \bar{q}_0$
1	0	38	0.019	58%	0.90907	0.90909	$+2 \times 10^{-5}$
1	1	40	0.019	65%	0.91008	0.91011	$+4 \times 10^{-5}$
2	0	40	0.016	62%	0.91959	0.91952	-7×10^{-5}
2	1	37	0.016	54%	0.92399	0.92392	-7×10^{-5}
3	0	69	0.017	48%	0.89264	0.89260	-4×10^{-5}
3	1	66	0.017	62%	0.89653	0.89651	-2×10^{-5}
4	0	38	0.017	66%	0.89865	0.89862	-3×10^{-5}
4	1	50	0.007	80%	0.89784	0.89781	-3×10^{-5}
5	0	188	0.008	84%	0.90161	0.90156	-5×10^{-5}
5	1	176	0.010	87%	0.90418	0.90413	-6×10^{-5}
6	0	127	0.007	83%	0.89619	0.89617	-2×10^{-5}
6	1	152	0.008	80%	0.89755	0.89753	-2×10^{-5}

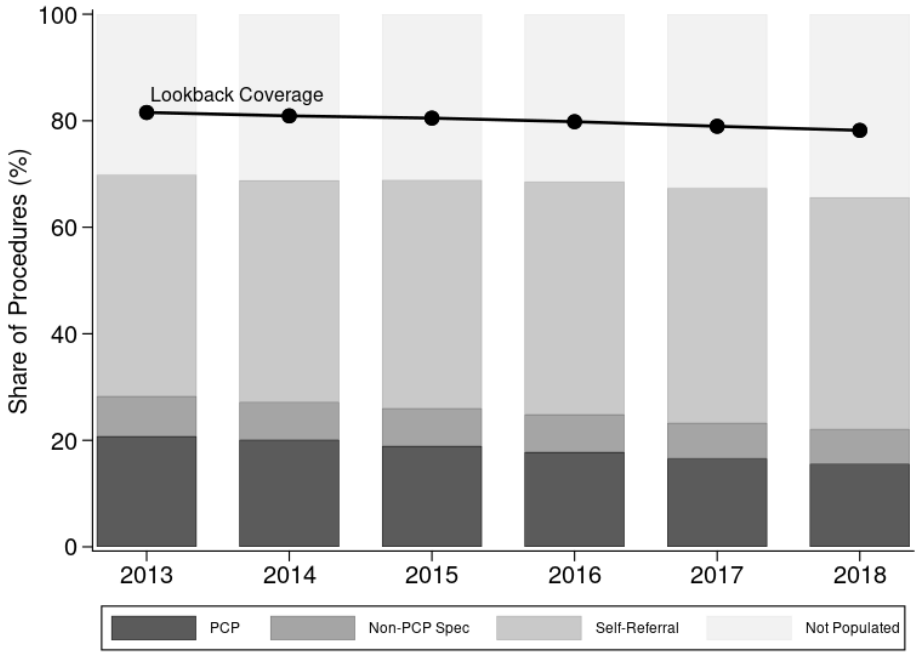
[a]Re-solution of the counterfactual fixed point from two extreme starting values (zero patients at every specialist and the observed patient volumes) for six test HRRs spanning small, medium, and large markets. "Per" denotes the time period (0 for 2013–2015, 1 for 2016–2018). "N spec" is the number of specialist-period observations in the HRR-period. "Mean $\lvert\Delta p_{ij}\rvert$" and "Share <1pp" describe the distribution of absolute differences in specialist-level predicted shares across the two starting values. "$\bar{q}_0$" and "$\bar{q}_1$" are the patient-weighted mean of specialist quality under each starting value.

Table 7: Example of Overlapping Windows[a]

			Window 6 (specialist 2)		Window 12 (specialist 3)		Window 17 (specialist 1)	
Referral	Specialist	Outcome	Position	Same	Position	Same	Position	Same
1	1	S	−5	0				
2	2	S	−4	1				
3	3	S	−3	0				
4	1	S	−2	0				
5	2	S	−1	1				
6	2	F	0					
7	3	S	1	0	−5	1		
8	3	S	2	0	−4	1		
9	1	S	3	0	−3	0		
10	3	S	4	0	−2	1		
11	1	S	5	0	−1	0		
12	3	S			0		−5	0
13	2	S			1	0	−4	0
14	1	S			2	0	−3	1
15	3	S			3	1	−2	0
16	2	S			4	0	−1	0
17	1	S			5	0	0	
18	3	S					1	0
19	2	S					2	0
20	1	S					3	1
21	3	S					4	0
22	2	S					5	0

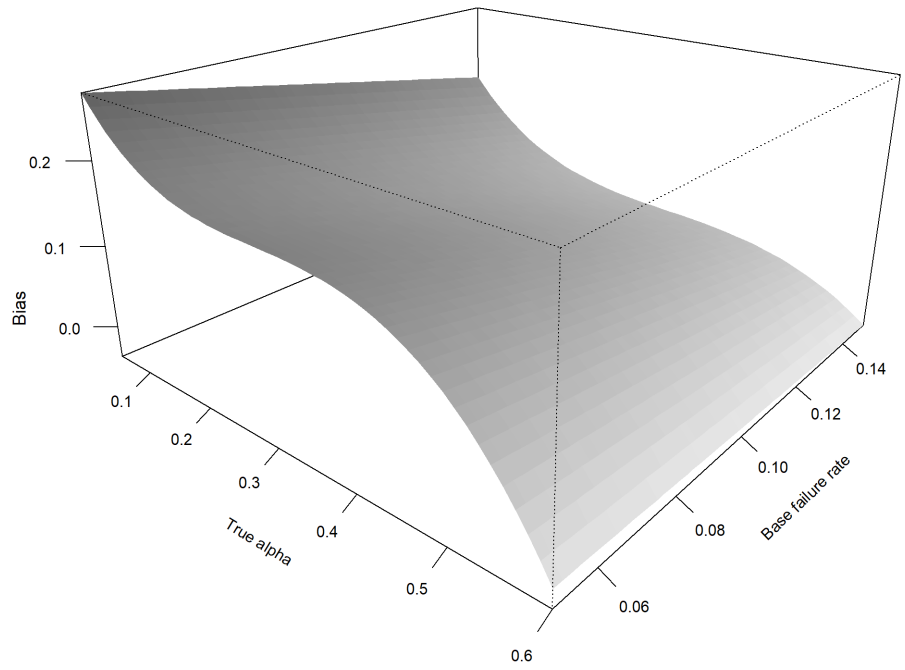
[a]One PCP's 22 referrals in order, with the specialist who received each and its surgical outcome (S for success, F for failure). Every referral defines a window, so these referrals define 22 windows; the table shows three of them. Each window is named for the referral at its position 0, and its specialist is the one who received that referral. Position is the referral's position in the window, and Same is one if the referral went to the window's specialist. Window 6 is a failure window; windows 12 and 17 are success windows. Referrals 7 through 11 are the post-period of window 6 and the pre-period of window 12.

Figure 1: Composition of the listed referring physician field and carrier-lookback coverage over time.[a]



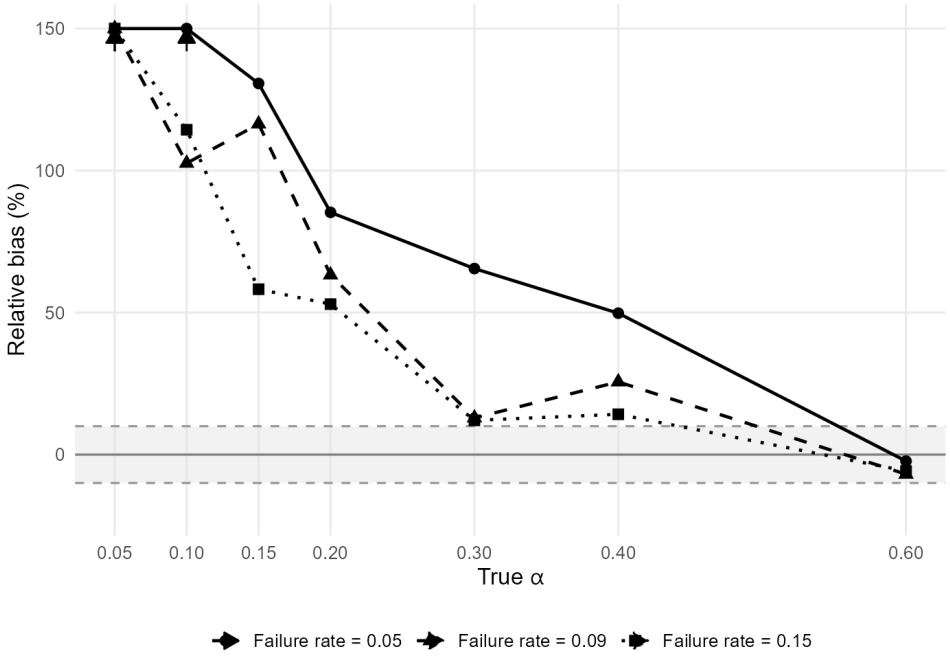
[a]Stacked bars show the composition of the `rfr_physn_npi` field on the surgeon's claim. The line shows the share of procedures for which our carrier-lookback method identifies a PCP.

Figure 2: Bias in $\hat{\alpha}$ as a function of the true α and the base failure rate, with quality dispersion fixed at 0.03.[a]



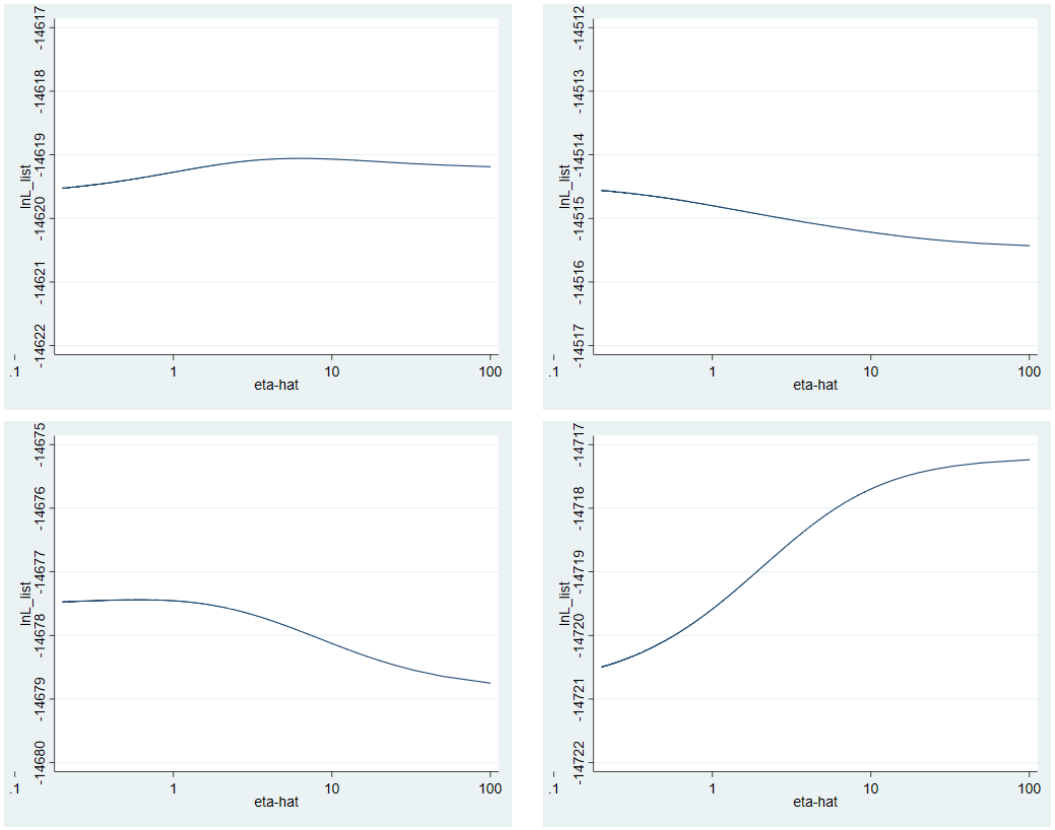
[a]The surface is smoothed using a polynomial fit to the simulation grid. Darker shading indicates larger bias.

Figure 3: Relative bias in $\hat{\alpha}$ (as a percentage of the true value) by base failure rate, with quality dispersion fixed at 0.03.[a]



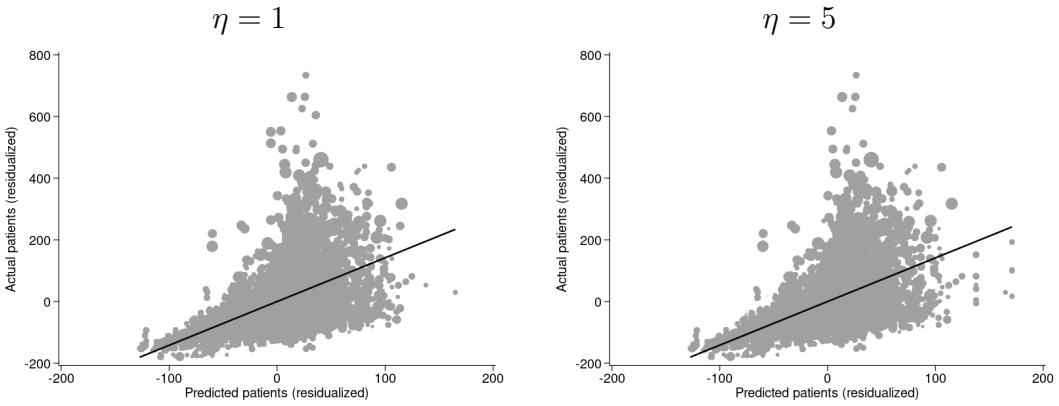
[a]The shaded band marks ±10% relative bias. Arrows indicate values exceeding 150% (truncated for readability).

Figure 4: Log-Likelihood Values in Four Example Simulations[a]



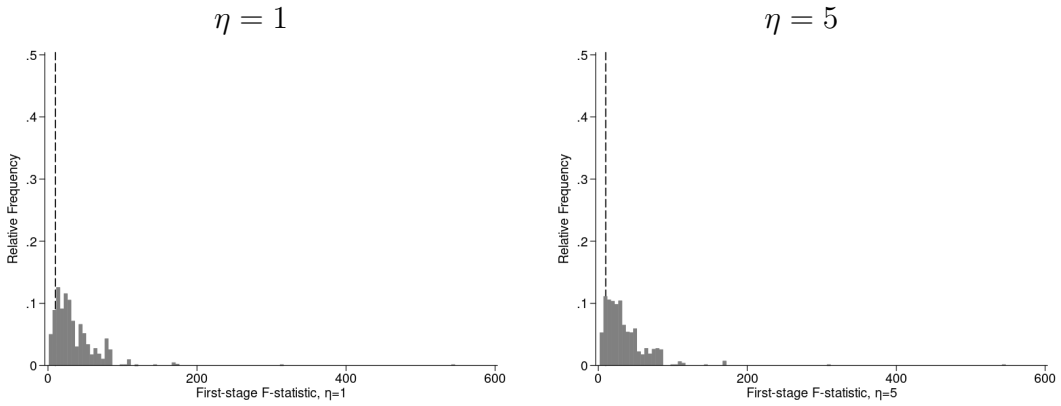
[a]Plots of the log-likelihood ("lnL_list") against the candidate value of η ("eta-hat," plotted on a log scale), from four example simulations. All other parameters are held at the values used to generate the simulated data, which consists of 2,500 patients and 25 specialists.

Figure 5: Partial scatter plots of actual versus predicted patient counts, residualized on HRR fixed effects and a time-period indicator.[a]



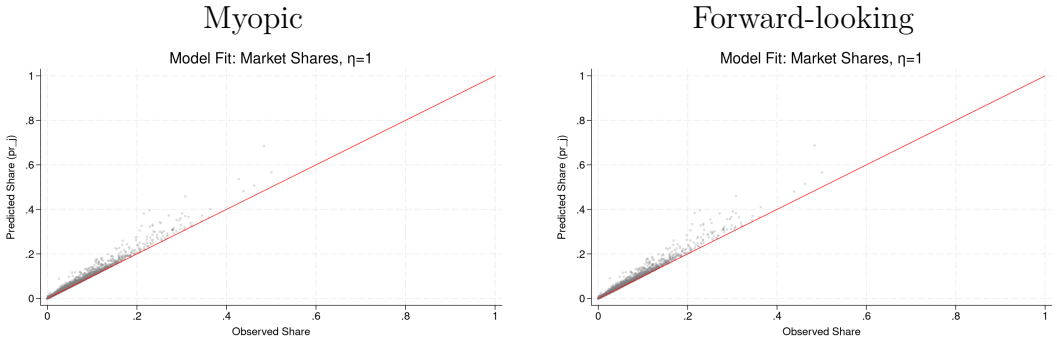
[a]Each point represents a specialist-time period observation, weighted by the inverse variance of the specialist fixed effect. The fitted line shows the first-stage relationship. Left panel shows $\eta = 1$, right panel shows $\eta = 5$.

Figure 6: Distribution of HRR-level first-stage F-statistics, weighted by patient volume.[a]



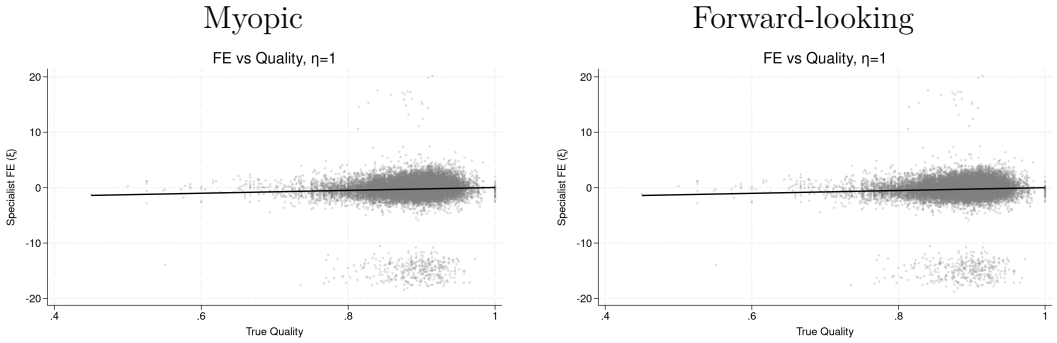
[a]Dashed line marks the conventional threshold of $F = 10$. Each HRR-level regression includes the distance-based instrument, a time-period indicator, and is weighted by the inverse variance of the specialist fixed effect. Left panel shows $\eta = 1$, right panel shows $\eta = 5$.

Figure 7: Predicted versus Actual Referral Shares ($\eta = 1$)[a]



[a]Each point is a specialist-period observation. The horizontal axis is the actual share of referrals the specialist receives in an HRR-period; the vertical axis is the model-predicted share. The 45-degree line is shown in black.

Figure 8: Specialist Fixed Effects and Observed Quality ($\eta = 1$)[a]



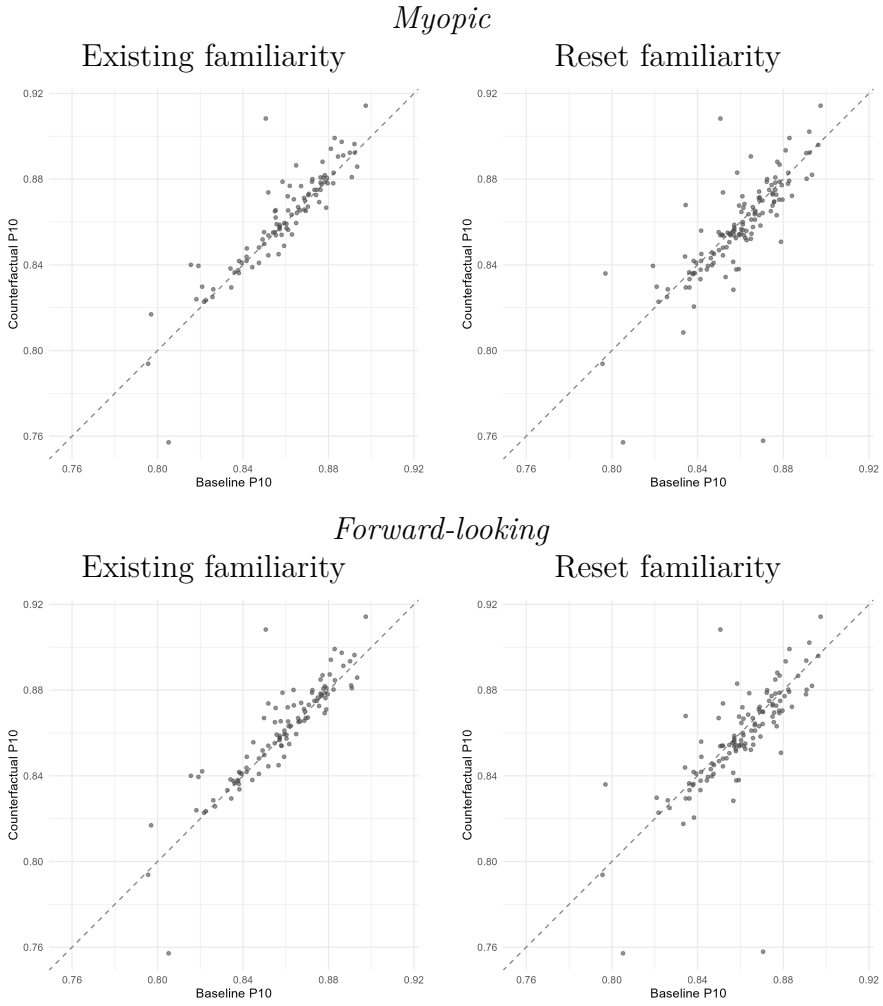
[a]Each point is a specialist. The horizontal axis is observed specialist quality (success rate); the vertical axis is the estimated specialist fixed effect ξ_j from the structural model. Within-HRR variation is the basis for the structural interpretation of α, so near-zero within-HRR correlation between ξ_j and quality supports the separate identification of the learning channel.

Figure 9: Counterfactuals from Full Information and No Familiarity[a]



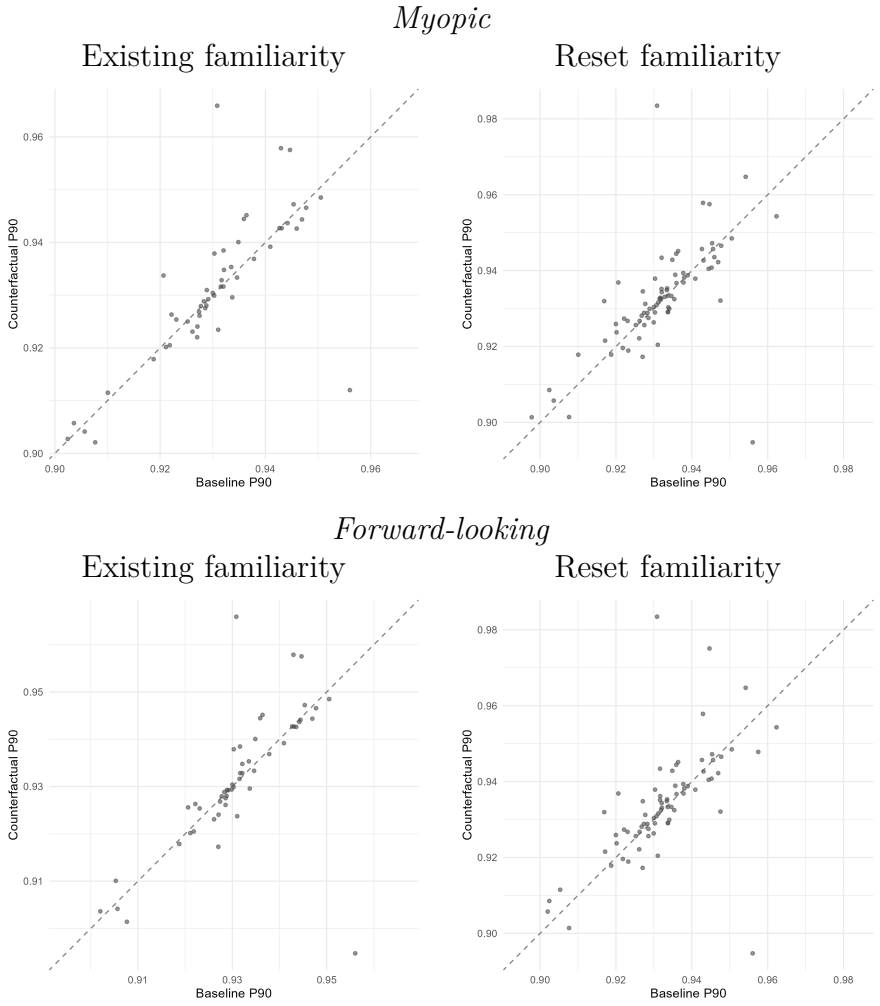
[a]Figure depicts the share of referrals reallocated and the resulting effects on patient health (i.e., the change in the probability of a successful surgery) under a counterfactual in which PCPs have full information about specialist quality and do not incorporate any familiarity in their referral decisions. Estimates are averaged across individuals within an HRR and presented as a histogram across HRRs. Results limited to HRRs in which our constrained maximization routine converged, 280 (277) HRRs in the myopic (forward-looking) case for $\eta = 1$ and 281 (280) in the myopic (forward-looking) case for $\eta = 5$.

Figure 10: Off-Diagonal Scatter of Patient-Weighted 10th-Percentile Quality ($\eta = 1$)[a]



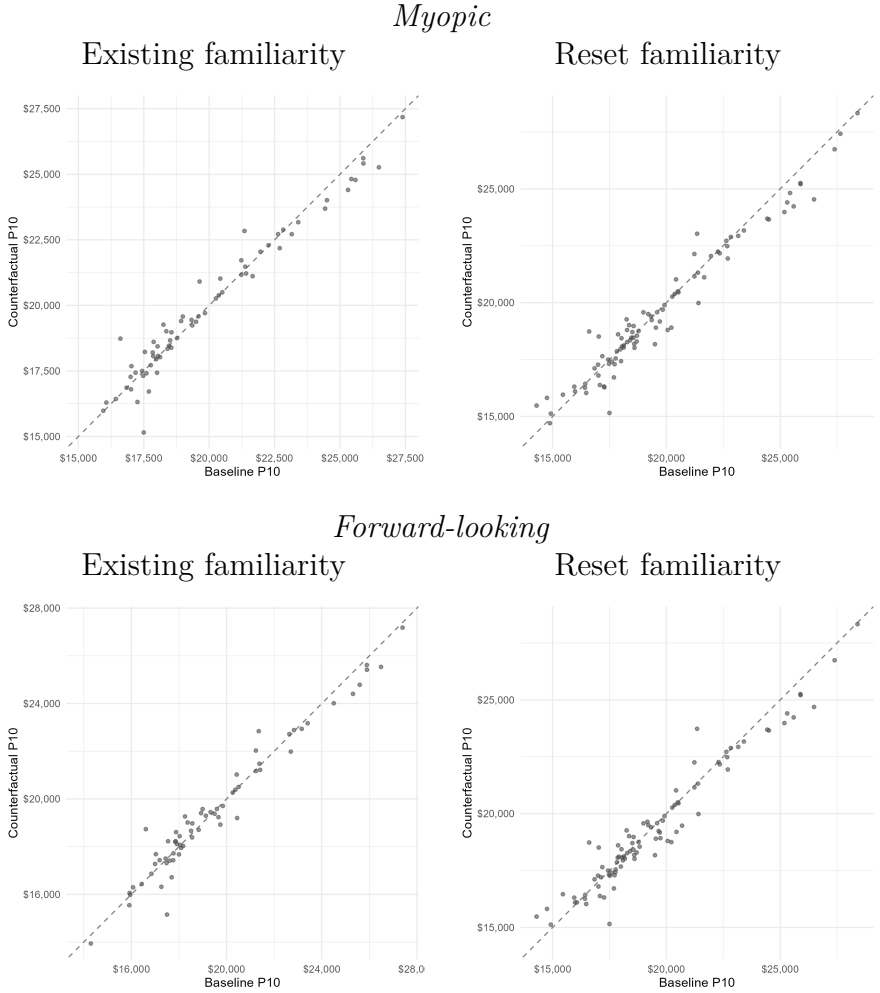
[a]Each point is an HRR where the patient-weighted 10th percentile of specialist quality under the counterfactual differs from the baseline value. HRRs with identical baseline and counterfactual percentile are dropped (they lie on the 45-degree line). The solid reference line is the 45-degree line.

Figure 11: Off-Diagonal Scatter of Patient-Weighted 90th-Percentile Quality ($\eta = 1$)[a]



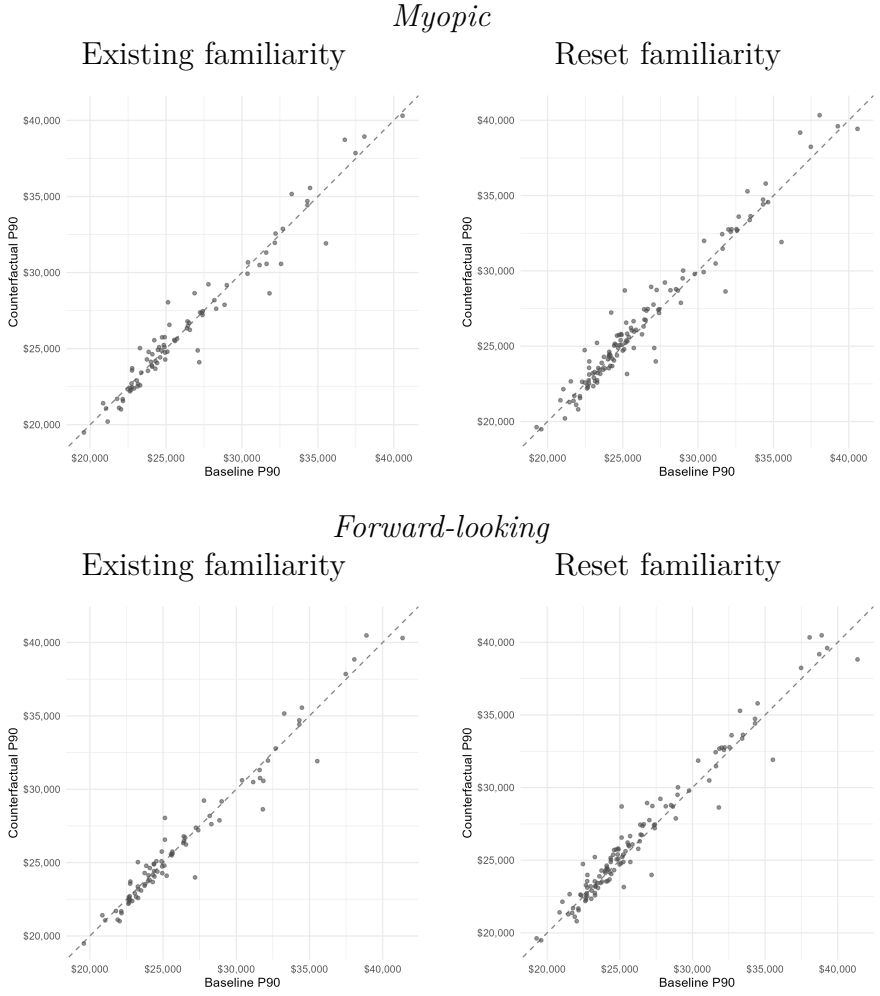
[a]Each point is an HRR where the patient-weighted 90th percentile of specialist quality under the counterfactual differs from the baseline value. HRRs with identical baseline and counterfactual percentile are dropped (they lie on the 45-degree line). The solid reference line is the 45-degree line.

Figure 12: Off-Diagonal Scatter of Patient-Weighted 10th-Percentile Spending ($\eta = 1$)[a]



[a]Each point is an HRR where the patient-weighted 10th percentile of episode spending under the counterfactual differs from the baseline value. HRRs with identical baseline and counterfactual percentile are dropped (they lie on the 45-degree line). The solid reference line is the 45-degree line.

Figure 13: Off-Diagonal Scatter of Patient-Weighted 90th-Percentile Spending ($\eta = 1$)[a]



[a]Each point is an HRR where the patient-weighted 90th percentile of episode spending under the counterfactual differs from the baseline value. HRRs with identical baseline and counterfactual percentile are dropped (they lie on the 45-degree line). The solid reference line is the 45-degree line.

Additional Figures

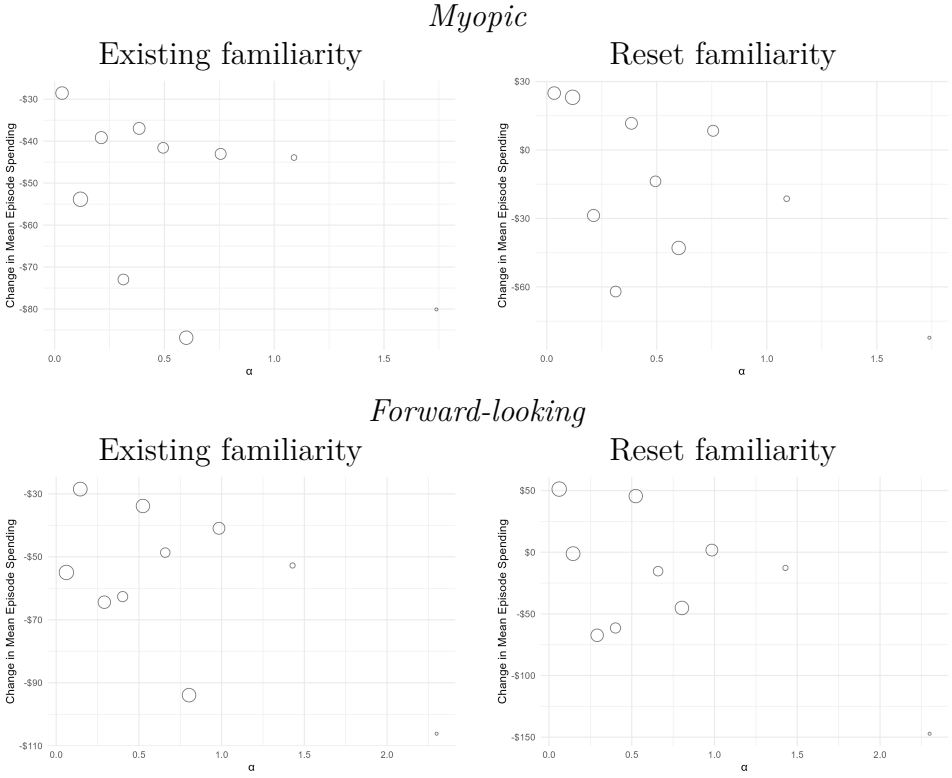
This section presents corresponding figures for $\eta = 5$ of the counterfactual gradients and off-diagonal percentile scatters shown elsewhere in the paper and appendix for $\eta = 1$.

Figure 14: Health Effects of Full Information by Estimated Learning Parameter $(\eta = 5)^a$



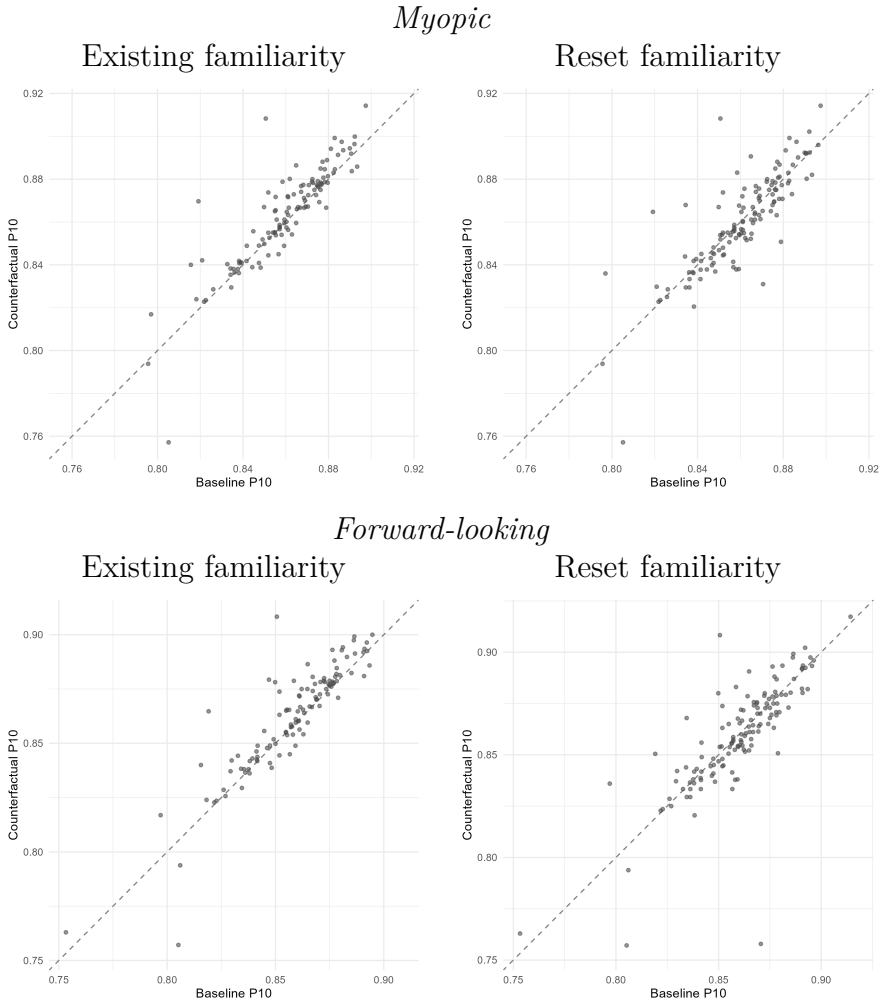
[a]Figure depicts the mean change in successes per 10,000 patients under each counterfactual, binned into ten patient-weighted quantiles of the estimated learning parameter $\hat{\alpha}$ within each HRR ($\hat{\alpha} > 0.001$). Marker size reflects the total patients in the bin.

Figure 15: Episode Spending Effects of Full Information by Estimated Learning Parameter ($\eta = 5$)[a]



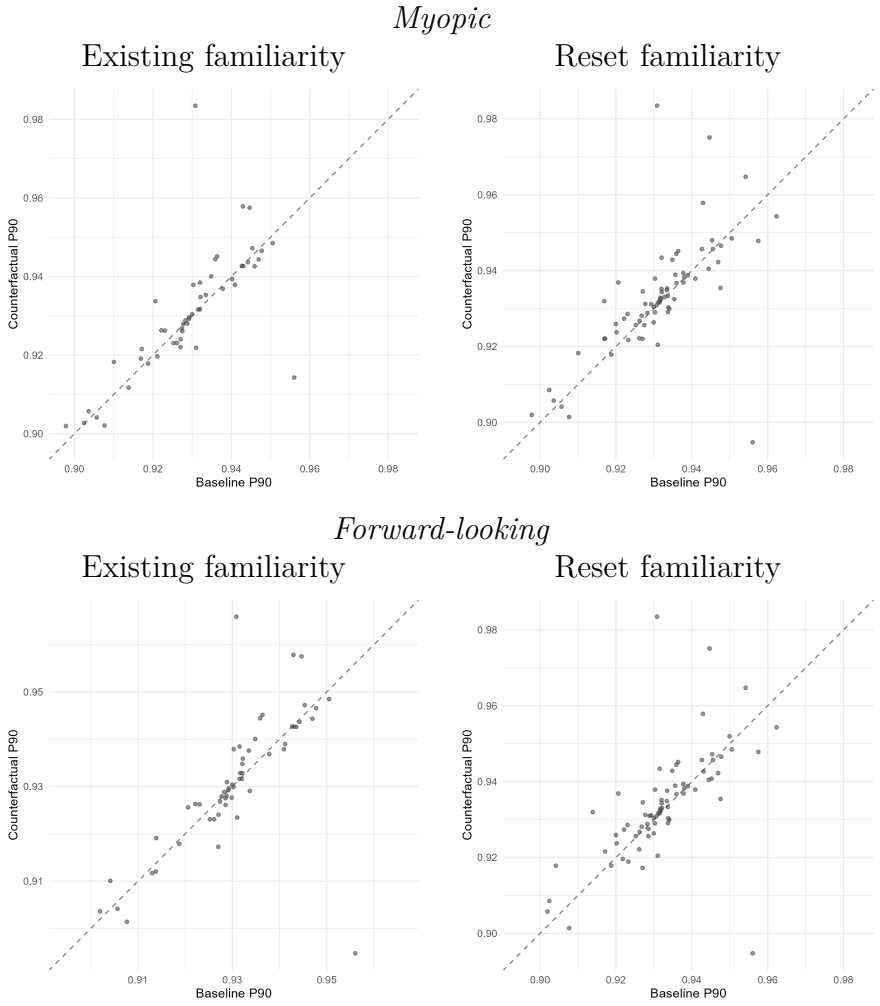
[a]Figure depicts the mean change in episode spending under each counterfactual, binned into ten patient-weighted quantiles of the estimated learning parameter $\hat{\alpha}$ within each HRR ($\hat{\alpha} > 0.001$). Marker size reflects the total patients in the bin.

Figure 16: Off-Diagonal Scatter of Patient-Weighted 10th-Percentile Quality ($\eta = 5$)[a]



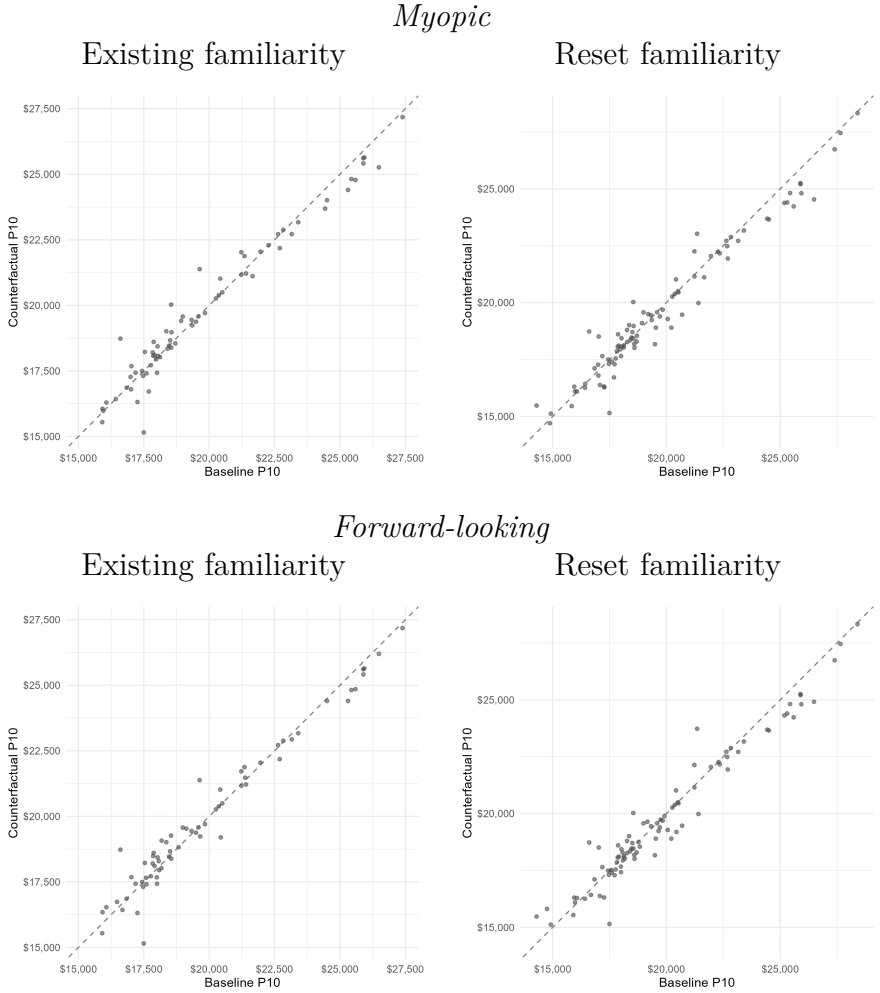
[a]Each point is an HRR where the patient-weighted 10th percentile of specialist quality under the counterfactual differs from the baseline value. HRRs with identical baseline and counterfactual percentile are dropped (they lie on the 45-degree line). The solid reference line is the 45-degree line.

Figure 17: Off-Diagonal Scatter of Patient-Weighted 90th-Percentile Quality ($\eta = 5$)[a]



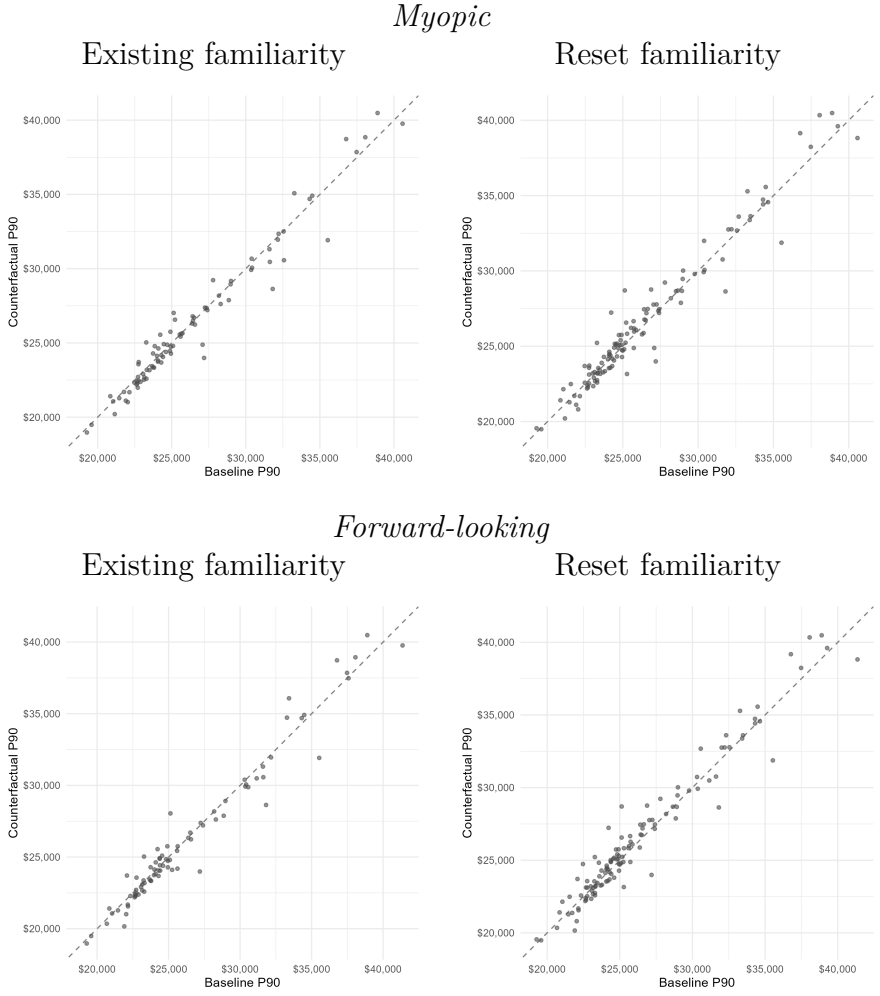
[a]Each point is an HRR where the patient-weighted 90th percentile of specialist quality under the counterfactual differs from the baseline value. HRRs with identical baseline and counterfactual percentile are dropped (they lie on the 45-degree line). The solid reference line is the 45-degree line.

Figure 18: Off-Diagonal Scatter of Patient-Weighted 10th-Percentile Spending ($\eta = 5$)[a]



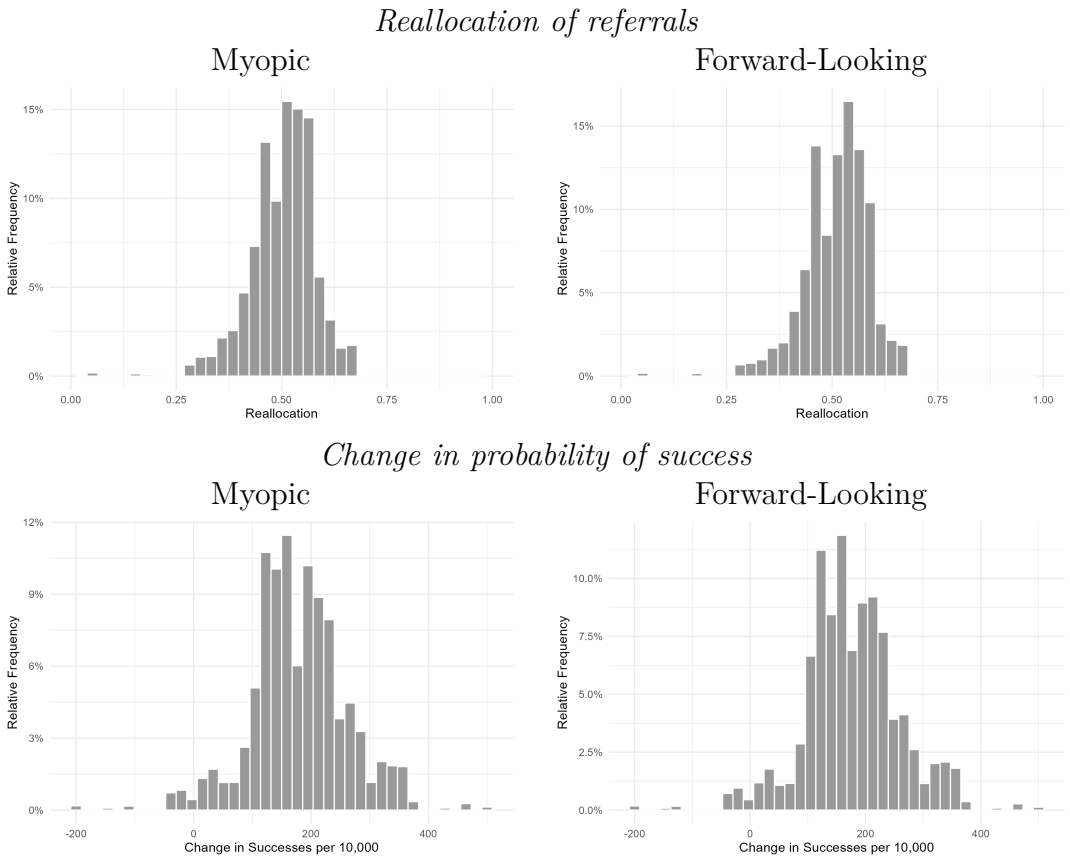
[a]Each point is an HRR where the patient-weighted 10th percentile of episode spending under the counterfactual differs from the baseline value. HRRs with identical baseline and counterfactual percentile are dropped (they lie on the 45-degree line). The solid reference line is the 45-degree line.

Figure 19: Off-Diagonal Scatter of Patient-Weighted 90th-Percentile Spending ($\eta = 5$)[a]



[a]Each point is an HRR where the patient-weighted 90th percentile of episode spending under the counterfactual differs from the baseline value. HRRs with identical baseline and counterfactual percentile are dropped (they lie on the 45-degree line). The solid reference line is the 45-degree line.

Figure 20: Counterfactual Changes with Quality-Maximizing Referrals ($\eta = 5$)[a]



[a]Figure depicts the share of referrals reallocated and the resulting change in probability of success under a counterfactual in which each PCP's belief is replaced with the true specialist success rate, specialist fixed effects and accumulated familiarity are set to zero, and α within each HRR is calibrated to match the larger of the distance or congestion utility components. Results are averaged across individuals within an HRR and presented as a histogram across HRRs.